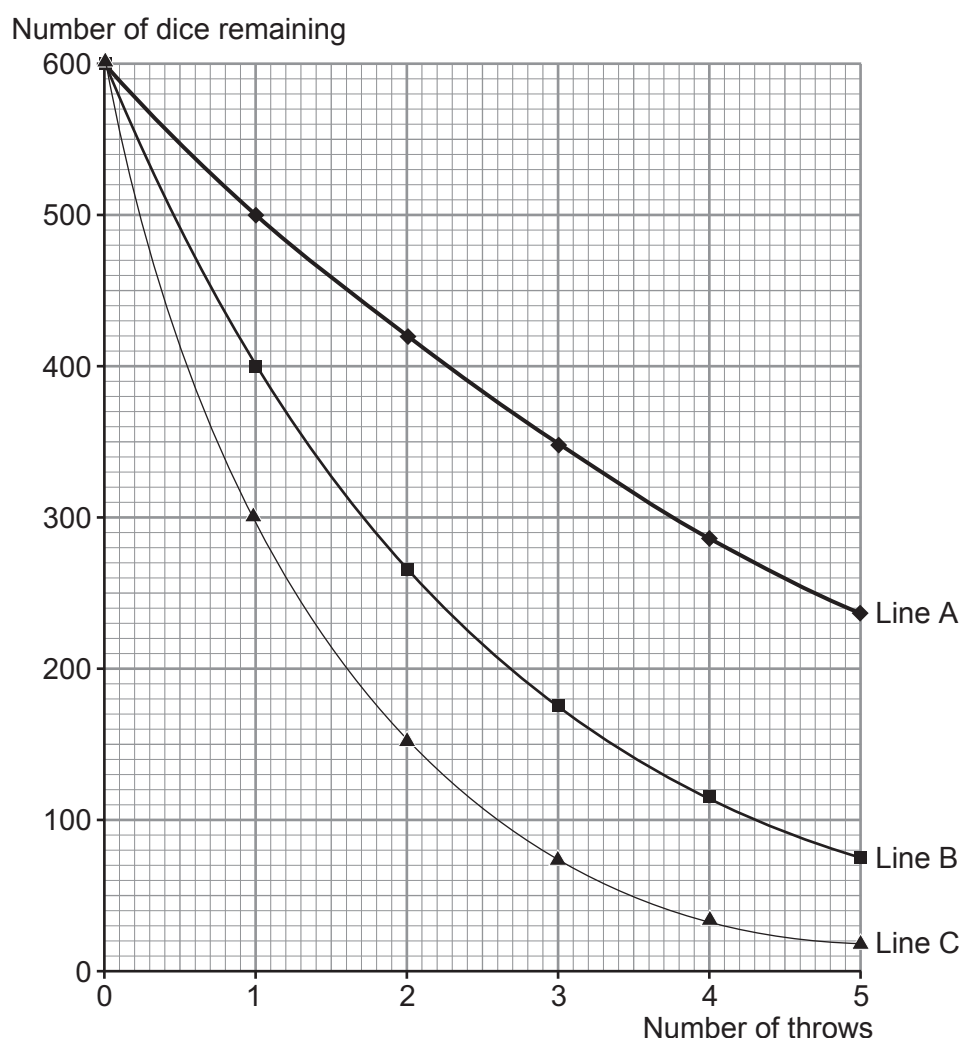


5. A physics teacher uses 600 identical red dice with her class to model the decay of different radioisotopes. To collect her first set of data she throws the 600 dice. She then removes all of the **sixes** that are face up. She then counts the remaining dice and repeats the process for a total of 5 throws.

In her second demonstration she repeats the experiment but removes all of the **fives** and **sixes** that are face up.

In her third demonstration, she removes all of the **fours**, **fives** and **sixes** that are face up.

She records her data on a spreadsheet. This quickly produces a graph for her class to see. The graph is shown below.



- (a) (i) Explain why the teacher chooses to use a large number of dice for her demonstration. [2]

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- (ii) Using only information from the graph tick (✓) the **three** correct statements. [3]
Space for working:

Line A represents the data from the third demonstration. ☐

Line B shows the equivalent of three half-lives in 5 throws. ☐

Line C could also represent the same demonstration modelled with 600 coins. ☐

Line A represents the radioisotope that decays at the quickest rate. ☐

Line B has an activity that is $\frac{1}{8}$ of its original after 5 throws. ☐

Line C represents the longest half-life. ☐

- (iii) The teacher suggests to her class that an improved model of radioactive decay would be to include 20 blue cubes, along with the 600 red dice, at the start. The blue cubes **would not** be removed during the demonstration but would be counted each time.

I. State what physical quantity the blue cubes represent. [1]

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II. State how the presence of the blue cubes affects **line A** on the graph. [1]

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(b) Strontium-93 (Sr-93) is a radioactive source which decays with a half-life of 8 minutes.

- (i) A sample of Sr-93 has a mass of 68.0 mg. Calculate the mass of undecayed Sr-93 left after 56 minutes. Show your working and give your answer to **1 significant figure**. [3]

Mass = mg

- (ii) A scientist works with Sr-93 which decays by beta and gamma emission. When he is not using the Sr-93, explain why, for his own safety, he has to store it in a lead box. [3]

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